

Wide Diameters of (n, k) -Star Graphs

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Abstract

An (n, k) -star graph $(S_{n,k})$ refers to a generalized version of an n -star graph (S_n) , where $S_{n,n-1}$ and S_n are isomorphic and $S_{n,1}$ is obviously a complete graph of n vertices (K_n) . Since Lin et al. (2004) already calculated the wide diameters in $(n, n-1)$ -star and $(n, 1)$ -star graphs, this study only considers an $S_{n,k}$ with $2 \leq k \leq n-2$. Lin et al. (2008) also computed upper and lower bounds of the wide diameter of an $S_{n,k}$. However, the upper bound is larger than the lower bound by one if $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$. Therefore, this work shortens the longest paths in the longest containers constructed by Lin et al. for $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n \geq 7$, such that the upper bound can be reduced at least one. Moreover, a better lower bound is provided for $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n < 7$. The wide diameter of an $S_{n,k}$ can then be determined since its upper and lower bounds are the same, and it is its diameter plus 1 for $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n \geq 7$, or plus 2 for otherwise.

1 Introduction

The topological structure of a massively parallel computer, an interconnection network or a distributed system can be modeled by a graph. Moreover, graph theory is a very powerful mathematical tool for designing and analyzing topological properties of interconnection networks such as permutation capability [3], Hamiltonian laceability [2], fault tolerance [12], [14], [15], [30], performance [5], reliability [31] and combinatorial properties [16]. Additionally, the wide diameter of an interconnection network (or a graph) can be computed by constructing vertex-disjoint paths, and have become increasingly important recently in studies of reliability, fault-tolerant ability, maximum parallelism and minimum transmission delay [7], [11], [17], [18], [21], [29].

Let G denote a graph. The vertex set and edge set of G are denoted by $V(G)$ and $E(G)$, respectively. Two vertices u and v are *adjacent* when they are joined by an edge e , where $u, v \in V(G)$ and $e \in E(G)$. All vertices adjacent to a vertex are its *neighbors*. The *distance* from vertex u to vertex v , represented by $d(G; u, v)$, refers to the length of a shortest path from u to v in G . The *diameter* of G , denoted by $d(G)$,

is defined as the maximum distance for all pairs of distinct vertices u and v in G [4]. A graph is *connected* when at least one path exists between any two vertices. The *connectivity* of a graph is defined as the minimum number of vertices whose removal renders it disconnected or trivial. Any two paths in a graph are *vertex-disjoint* to each other if they have no common vertex in each path.

A set of vertex-disjoint paths between vertices u and v , denoted by $C(G; u, v)$, is an (u, v) -container in G . The number of paths in $C(G; u, v)$ is the *width* of $C(G; u, v)$. An (u, v) -container with width ω is denoted by $C_\omega(G; u, v)$. The length of a $C(G; u, v)$ ($C_\omega(G; u, v)$), denoted by $l(C(G; u, v))$ ($l(C_\omega(G; u, v))$), is defined as the length of the longest path in $C(G; u, v)$ ($C_\omega(G; u, v)$) [21]. Let κ be the connectivity of G . According to Menger's theorem, at least κ vertex-disjoint paths exist between any two distinct vertices in G [4]. An (u, v) -container with width κ is abbreviated as *container*. A *best-container* between u and v , denoted by $C_\kappa^*(G; u, v)$, is the container among all $C_\kappa(G; u, v)$ s, with the shortest length. Let $d_\kappa(G; u, v)$ indicate the κ -wide distance (wide distance for short) from u to v ; thus, $d_\kappa(G; u, v) = l(C_\kappa^*(G; u, v))$ [21]. The κ -wide diameter (wide diameter for short) of G , denoted by $d_\kappa(G)$, is defined as the maximum of $d_\kappa(G; u, v)$ s for all pairs of distinct vertices u and v in G .

An (n, k) -star graph, denoted by $S_{n,k}$, is a generalized version of an S_n . The structure of an $S_{n,k}$ is formally defined in Section 2. Some basic properties of an $S_{n,k}$, such as diameter [9], connectivity [9], average distance [10], Hamiltonicity [22], spanning connectivity [23], broadcasting [8], embedding [6], optimal spanning tree [25], lower bound of feedback vertex sets [29], have recently been computed or derived. These measurement results demonstrate that an $S_{n,k}$ has excellent topological properties.

Lin et al. proposed an one-to-one routing algorithm for constructing a container of width $n-1$ between any pair of vertices in an $S_{n,k}$ with connectivity $n-1$. An upper bound of the wide diameter of an $S_{n,k}$ is derived, and it is $d(S_{n,k})+2$ [26]. They also demonstrated that the lower bound of $d_\kappa(G)$ for a regular graph G with connectivity $\kappa \geq 2$ is $d(G)+1$ and $d_{n-1}(S_{n,k}) \geq d(S_{n,k})+2$ for $2 \leq k \leq \lfloor n/2 \rfloor$. Measurement results indicate that $d_{n-1}(S_{n,k})$ is $d(S_{n,k})+2$ for $2 \leq k \leq \lfloor n/2 \rfloor$ or between $d(S_{n,k})+1$ and

$d(S_{n,k})+2$, for $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ [26]. To reduce the upper bound of $d_{n-1}(S_{n,k})$, this work shortens the longest paths in the longest containers constructed by Lin *et al.* for $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n \geq 7$. Moreover, a better lower bound is also provided for $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n < 7$. Thus, $d_{n-1}(S_{n,k})$ can then be determined since its upper and lower bounds are the same, and $d_{n-1}(S_{n,k}) = d(S_{n,k}) + 1$ for $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n \geq 7$, or $d_{n-1}(S_{n,k}) = d(S_{n,k}) + 2$ for otherwise.

The remainder of this paper is organized as follows. Section 2 formally describes the topological structure of an $S_{n,k}$. Some notations and basic properties of an $S_{n,k}$ are also introduced and discussed. Section 3 calculates the wide diameter of an $S_{n,k}$. Conclusions are finally drawn in Section 4.

2 Background

An $S_{n,k}$ refers to a generalized version of an S_n , where $S_{n,n-1}$ and S_n are isomorphic [9] and $S_{n,1}$ is obviously a complete graph of n vertices (K_n). An S_n is a Cayley graph with a regular and hierarchical structure, and has been recognized as an attractive alternative to an n -cube for building massively parallel computers [1], [13]. An S_n has significant advantages, including low degree, small diameter, scalability and symmetry [1]. Additionally, many advantageous properties of an S_n , such as parallel paths [7], [18], spanning trees [27] and a high degree of fault tolerance [19], [20], [24], [28], have been investigated. However, an S_n has a rigorous restriction, i.e., the number of vertices must satisfy $n!$. Thus, additional $(n-1) \times (n-1)!$ vertices are required to expand an S_{n-1} to an S_n . To eliminate this restriction, the new graph $S_{n,k}$ has been presented [9]. An $S_{n,k}$ retains many of the attractive properties of an S_n , such as vertex symmetry, hierarchical structure, maximal fault tolerance and simple shortest-path routing, but only additional vertices of $(n-1)(n-1)!/(n-k)!$ are utilized to expand an $S_{n-1,k-1}$ to an $S_{n,k}$.

An $S_{n,k}$ is an undirected-regular and vertex-symmetrical graph. Parameters n and k , where $1 \leq k \leq n-1$, are applied to determine the topological properties, such as the number of vertices in an $S_{n,k}$ is $n!/(n-k)!$, the connectivity of an $S_{n,k}$ is $n-1$ and the diameter of an $S_{n,k}$ is $2k-1$ if $1 \leq k \leq \lfloor n/2 \rfloor$ or $k + \lfloor (n-1)/2 \rfloor$ if $\lfloor n/2 \rfloor + 1 \leq k \leq n-1$ [9]. A vertex ρ in an $S_{n,k}$ can be represented as a label $p_1 p_2 \dots p_k$. The vertex set $V(S_{n,k})$ is defined as $\{p_1 p_2 \dots p_k \mid p_i \neq p_j, \text{ for } i \neq j \text{ and } p_i, p_j \text{ in } \langle n \rangle\}$. Each element in $\langle n \rangle$ is referred to as a *symbol*. Symbol $\in \langle k \rangle$ ($\in \langle n \rangle - \langle k \rangle$) is an *internal* (*external*) symbol. Moreover, symbol p_i in label $p_1 p_2 \dots p_k$ is as follows: (1) *fixed* if $p_i = i$ (notably, position i is the *desired position* for symbol i); (2) *misplaced* if $p_i \neq i$ and p_i is an internal symbol, and (3) *inner external* if p_i is an external symbol. A symbol

not in the label $p_1 p_2 \dots p_k$ is *desired* if it is an internal symbol, and *outer external* if it is an external symbol. For instance, with $\rho = 369482$ in $S_{9,6}$, $p_4 = 4$ is a fixed symbol. Additionally, $p_1 = 3$, $p_2 = 6$ and $p_6 = 2$ are misplaced symbols; symbols 1 and 5 are desired symbols; $p_3 = 9$ and $p_5 = 8$ are inner external symbols, and symbol 7 is an outer external symbol. Vertex ρ is adjacent to ρ_{p_i} and ρ_x by the following adjacent rules.

1. $\rho_{p_i} = p_i p_2 \dots p_{i-1} p_1 p_{i+1} \dots p_k$, for all $2 \leq i \leq k$ (swapping p_1 with p_i), and
2. $\rho_x = x p_2 \dots p_k$, for all x is a desired or outer external symbol (replacing p_1 with x).

The edge connecting ρ and ρ_{p_i} is referred to as i -edge, and ρ_{p_i} is referred to as i -neighbor of ρ . The edge connecting ρ and ρ_x is referred to as 1-edge, and ρ_x is referred to as 1-neighbor of ρ . Figure 1 shows the structure of an $S_{4,2}$.

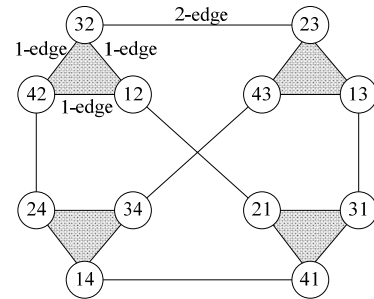


Figure 1. The structure of an $S_{4,2}$.

The degrees of all vertices are $n-1$; each vertex is connected with $(n-k)$ 1-edges, and an i -edge for each i , $2 \leq i \leq k$. Figure 2 illustrates all neighbors of vertex 1234 in $S_{7,4}$.

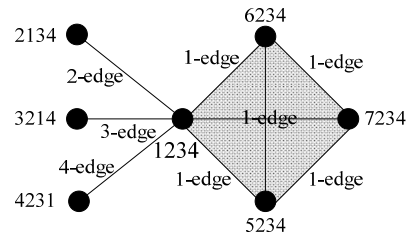


Figure 2. All neighbors of vertex 1234 in an $S_{7,4}$.

For the label of vertex ρ in an $S_{n,k}$, all non-fixed symbols are represented as a *cycle representation*. The cycle representation is similar to the cycle permutation for an S_n . Restated, when $p_x = y$, $p_y = z$, and $p_z = x$, these non-fixed symbols can be presented as a cycle $(x \ y \ z)$, where x denotes the cycle *head*, indicating that the desired position of a symbol is occupied by the next symbol in the cycle. Notably, cyclically shifting the symbols in one cycle does not alter the occupying property of each symbol. Additionally, a label does not adopt the positions from $k+1$ to n , which are called *pseudo-positions*.

For instance, vertex $\rho = 369482$ in $S_{9,6}$ can be extended to 369482751, in which symbols 7, 5 and 1 are occupied at pseudo-positions 7, 8 and 9,

respectively, as indicated by cycles (1 3 9)(5 8)(2 6). Cycle (2 6) is an *internal cycle*, since all symbols are internal. Cycles (1 3 9) and (5 8) are *external cycles*, since an external symbol exists in each of them. However, all individual external cycles (1 3 9) and (5 8) should be merged into one as (1 3 9 5 8). This indicates that pseudo-position 9 should be occupied by the desired symbol 5, and pseudo-position 8 should be occupied by the desired symbol 1. Restated, 369482 is extended as 369482715 instead of 369482751; in addition, the cycle representation of ρ is (1 3 9 5 8)(2 6).

Let m and e be the number of misplaced and inner external symbols of vertex ρ , respectively. The cycle representation comprises $m+2e$ symbols in total and implies that $m+2e$ symbols are not in their desired positions. Interestingly, Chiang and Chen showed in 1995 that $d(S_{n,k}; \rho, \varepsilon)$ is $c+m+2e$ if $p_1=1$ or $c+m+2e-2$ if $p_1 \neq 1$, where c denotes the number of cycles when all external cycles are merged into one. Moreover, they also revealed that $d(S_{n,k})=2k-1$ if $1 \leq k \leq \lfloor n/2 \rfloor$ or $k + \lfloor (n-1)/2 \rfloor$ if $\lfloor n/2 \rfloor + 1 \leq k \leq n-1$. This work adopts $d(S_{n,k})$ to represent the length of each of the constructed paths in an $S_{n,k}$.

The vertex $\varepsilon=12\dots k$ is an *identity vertex* whose $p_i=i$ for all $1 \leq i \leq k$. Due to vertex symmetry of an $S_{n,k}$, any vertex can be mapped to the identity vertex ε by a one-to-one mapping function by renaming the symbols. Therefore, without loss of generality, considering the routing problem between any two vertices in an $S_{n,k}$, one of them is always assumed to be the identity vertex ε .

To construct a path π from ρ to ε in an $S_{n,k}$, all misplaced, desired and inner external symbols of ρ should be corrected one by one to transform the label ρ into label ε . The following routing rules are applied to correct symbol s . Each rule can join 1 or 2 edges (vertices) in a path.

R1. If symbol s is at position 1, then

- If s is an internal symbol, then move it to its desired position. (The symbol residing at position s is automatically moved to position 1 when s is corrected.)
- If s is an external symbol, then replace s with a desired symbol. (Symbol s is removed from the label.)

R2. If symbol s is not at position 1, then

- Exchange s with the symbol in position 1, then apply rule R1.

One edge contributes to path π if R1 is applied, and two edges contribute to path π if R2 is applied. Clearly, all symbols can be corrected in different orders, such that different paths from ρ to ε can be constructed.

Assuming that vertex ρ 's cycle representation is $C_1 C_2 \dots C_c$, let $C_i = (s_{i,1} s_{i,2} \dots s_{i,k_i})$ and the length of C_i be k_i . Symbol $s_{i,x}$ is then considered the x th symbol in C_i , thus, $s_{i,1}$ is the head of C_i . In particular, if a cycle

contains symbol 1, then assume the cycle is C_1 , and regularize symbol 1 as s_{1,k_1} and p_1 as the head $s_{1,1}$, by rotating C_1 . A *symbol sequence* (sequence for short) is defined as $\sigma(s_{1,1}) = \langle C_1, C_2, \dots, C_c \rangle = \langle s_{1,1}, s_{1,2}, \dots, s_{1,k_1}, s_{2,1}, s_{2,2}, \dots, s_{2,k_2}, \dots, s_{c,k_c} \rangle$. For convenience, $\sigma(s_{1,1})$ is also renumbered as $\sigma(s_1) = \langle s_1, s_2, \dots, s_{m+2e} \rangle$, where $k_1 + k_2 + \dots + k_c = m+2e$. These two expressions are used alternately throughout the remainder of this study. For example, let $\rho=369482$ in $S_{9,6}$. The cycle representation of ρ is (1 3 9 5 8)(2 6), and sequence $\sigma(s_{1,1})$ is arranged as $\langle C_1, C_2 \rangle = \langle 3, 9, 5, 8, 1, 2, 6 \rangle$, where C_1 is rotated as (3 9 5 8 1) and C_2 is (2 6).

If path $\pi(s_1)$ is constructed in the order $\sigma(s_1)$, then each of $m+2e$ symbols should be corrected by applying R1 or R2. If $p_1=1$, then none of the c heads is at position 1 when these heads turn to be corrected and R2 is applied. Conversely, each of the remaining $m+2e-c$ symbols in $\sigma(s_1)$ can be corrected by applying R1, since the symbol can be automatically located at position 1 as a result of the correction of its previous symbol in $\sigma(s_1)$. Thus, $2c+(m+2e-c) = c+m+2e$ edges contribute to the constructed path $\pi(s_1)$. If $p_1 \neq 1$, then all c heads except $s_{1,1}$ are not at position 1, and s_{1,k_1} no longer needs to be corrected, since position 1 is its desired position, and $2(c-1)+(m+2e-(c-1)-1) = c+m+2e-2$ edges would contribute to the constructed path $\pi(s_1)$. Let $l(\pi(s_1))$ denote the length of path $\pi(s_1)$. Therefore, path $\pi(s_1)$ is a shortest path, and $l(\pi(s_1))=d(S_{n,k}; \rho, \varepsilon)$.

Lemma 2.1 [26]. Assuming that π_1 and π_2 are two distinct paths from ρ to ε constructed by R1 and/or R2, the corrections of symbols a and i (b and j) are the first (last) corrections in π_1 and π_2 , respectively. If the correction order of b is prior to a in π_2 then π_1 and π_2 are vertex-disjoint except for ρ and ε .

Corollary 2.2[26]. Paths π_1 and π_2 are two paths constructed by R1 and/or R2. Assuming π_1 is a path from vertex u to vertex v , and the correction of symbol a (symbol b) is the first (last) correction in π_1 . If the correction order of b is prior to a in π_2 then π_1 and π_2 are vertex-disjoint except for u, v .

It is known that $d(S_{n,k}; \rho, \varepsilon) \leq d(S_{n,k})$, and can be represented as $d(S_{n,k}) - \Delta$, where Δ is a non-negative integer. The formula for Δ for determining the $l(C_{n-1}(S_{n,k}; \rho, \varepsilon))$ is derived in the following.

An internal cycle contains at least two internal symbols, and one desired symbol collocating one inner external symbol can form an external cycle. Therefore, other misplaced symbols in a cycle, which have no effect on the number of cycle c , are termed *redundant symbols*. For example, in vertex (2 3 4)(5 6 7) in $S_{9,6}$, both internal cycle (2 3 4) and external cycle (5 6 7) have one redundant symbol. Let m' denote the number of redundant symbols in all cycles, and x be the number of internal cycles in the cycle representation. Undoubtedly, $m'=m-2x$, and m' is

even (or odd) if m is even (or odd). Consequently, $c=(m-m')/2+1$ ($c=(m-m')/2$) if $e \neq 0$ ($e=0$).

As $d(S_{n,k}; \rho, \varepsilon) \leq d(S_{n,k})$, $d(S_{n,k}; \rho, \varepsilon)$ can be represented as $d(S_{n,k}) - \Delta$, where $\Delta = d(S_{n,k}) - d(S_{n,k}; \rho, \varepsilon)$ is called a *difference*. For vertex ρ in a specific $S_{n,k}$, since $d(S_{n,k})$ is invariant, Δ is only related to $d(S_{n,k}; \rho, \varepsilon)$. Additionally, Δ is denoted as Δ_1 if $2 \leq k \leq \lfloor n/2 \rfloor$, Δ_2 if $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and n is even, or as Δ_3 if $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and n is odd. If ρ contains f fixed symbols other than symbol 1, and contains e' outer external symbols, then Δ_1 , Δ_2 , and Δ_3 are derived in the following lemmas for further computation.

Lemma 2.3 [26]. When $e > 0$, $\Delta_1 = (m+m'+4f)/2$, $\Delta_2 = (m'+e'+3f-1)/2$ if $p_1=1$ or $\Delta_2 = (m'+e'+3f)/2$ if $p_1 \neq 1$, and $\Delta_3 = (m'+e'+3f)/2$ if $p_1=1$ or $\Delta_3 = (m'+e'+3f+1)/2$ if $p_1 \neq 1$.

Lemma 2.4 [26]. When $e=0$, $\Delta_1 = (m+m'+4f)/2+1$, $\Delta_2 = (m'+e'+3f-1)/2+1$ if $p_1=1$ or $\Delta_2 = (m'+e'+3f)/2+1$ if $p_1 \neq 1$, and $\Delta_3 = (m'+e'+3f)/2+1$ if $p_1=1$ or $\Delta_3 = (m'+e'+3f+1)/2+1$ if $p_1 \neq 1$.

Corollary 2.5 [26]. Assuming $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and n is even (odd), $m'+e'+3f$ is odd (even) if $p_1=1$ or $m'+e'+3f$ is even (odd) if $p_1 \neq 1$.

3 Wide Diameter

This section first shows that $d_{n-1}(S_{n,k}) \leq d(S_{n,k}) + 2$. Second, $d_k(G) \geq d(G) + 1$ if G is a regular graph and $\kappa \geq 2$, and $d_{n-1}(S_{n,k}) \geq d(S_{n,k}) + 2$ are shown by constructing a best-container between a specific vertex pair u and v in an $S_{n,k}$ for $2 \leq k \leq \lfloor n/2 \rfloor$ or $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$, $n < 7$. Third, if $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$, $n \geq 7$, some refinements are used for some containers with maximum distance $d(S_{n,k}) + 2$ to shorten its longest path in order to obtain containers of length at most $d(S_{n,k}) + 1$. Therefore, $d_{n-1}(S_{n,k})$ can be then determined since its upper bound and lower bound are the same, and $d_{n-1}(S_{n,k}) = d(S_{n,k}) + 1$ for $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n \geq 7$, or $d_{n-1}(S_{n,k}) = d(S_{n,k}) + 2$ for otherwise.

3.1 One-to-One Vertex-Disjoint Paths

The routing algorithm proposed by Lin *et al.* is presented in this subsection. The constructed $n-1$ vertex-disjoint paths are divided into three groups, i.e., α -paths, β -paths and γ -paths and are shown in Lemmas 3.1, 3.3 and 3.5, respectively. Moreover, the length of each constructed path is computed in Lemmas 3.2, 3.4, and 3.6, respectively. Each α -path established by correcting the symbol sequence of vertex ρ in order. Each β -path is built by appending a fixed symbol into the symbol sequence, and not correcting it until all other symbols are corrected. Similarly, each γ -path is built by appending an outer external symbol into the symbol sequence, and not

correcting it until all other symbols are corrected. The constructed paths of these three groups are described in the following lemmas.

Lemma 3.1 [26]. $m+2e$ ($m+2e-1$) vertex-disjoint α -paths are constructed if $p_1=1$ ($p_1 \neq 1$).

Lemma 3.2 [26]. The lengths of the α -paths are as follows:

- (1) If $p_1=1$, then the $m+2e$ paths are at most $d(S_{n,k})$ long.
- (2) If $p_1 \neq 1$, then c paths are at most $d(S_{n,k})$ long, and $m+2e-c-1$ paths are at most $d(S_{n,k})+1$ ($d(S_{n,k})+2$) long when $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and n is odd (when $2 \leq k \leq \lfloor n/2 \rfloor$ or $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and n is even).

Lemma 3.3 [26]. $k-m-e-1$ ($k-m-e$) vertex-disjoint β -paths are constructed if $p_1=1$ ($p_1 \neq 1$). Each $\pi(\beta_i)$ is vertex-disjoint to all $\pi(s_\alpha)$ s.

Lemma 3.4 [26]. The lengths of β -paths are as follows.

- (1) If $p_1=1$, then the length of a β -path is at most $d(S_{n,k})+1$ when $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and n is even, or $d(S_{n,k})$ otherwise.
- (2) If $p_1 \neq 1$, then the length of a β -path is at most $d(S_{n,k})+2$.

Lemma 3.5 [26]. $n-k-e$ vertex-disjoint γ -paths are constructed; $\pi(\gamma_i)$ is vertex-disjoint to all $\pi(s_\alpha)$ s and $\pi(\beta_i)$ s.

Lemma 3.6 [26]. The length of each γ -path is at most $d(S_{n,k})+2$.

Lemmas 3.1 to 3.6 indicate that $l(C_{n-1}(S_{n,k}; \rho, \varepsilon))$ is at most $d(S_{n,k})+2$, and can serve as an upper bound of $d_{n-1}(S_{n,k})$.

Lemma 3.7 [26]. $d_{n-1}(S_{n,k}) \leq d(S_{n,k}) + 2$.

3.2 Wide Diameter

By Lemma 3.7, an upper bound of $d_{n-1}(S_{n,k})$ is $d(S_{n,k})+2$ can be obtained. This subsection first computes the lower bound of a regular graph G with connectivity $\kappa \geq 2$ and it is diameter plus 1. Next, determine that a lower bound of $d_{n-1}(S_{n,k})$ is $d(S_{n,k})+2$ for $2 \leq k \leq \lfloor n/2 \rfloor$, or $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n < 7$. Finally, for an $S_{n,k}$ with $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$, $n \geq 7$, the longest path of the constructed container with maximum distance $d(S_{n,k})+2$ is shorten by some refinements and then obtained that an upper bound of $d_{n-1}(S_{n,k})$ is $d(S_{n,k})+1$ for $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$, $n \geq 7$. These results demonstrated that the upper bound and lower bound of an $S_{n,k}$, $2 \leq k \leq n-2$ are the same. Therefore, the wide diameter of an $S_{n,k}$ is determined.

3.2.1 Lower Bound

This subsection first shows that the lower bound of

the wide diameter of a regular graph G with connectivity $\kappa \geq 2$. The lower bound of the wide diameter of an $S_{n,k}$ is then computed.

Lemma 3.8 [26]. If G is a regular graph with connectivity $\kappa \geq 2$, then $d_\kappa(G) \geq d(G) + 1$.

Since an $S_{n,k}$ is a regular graph with connectivity $\kappa = n - 1 \geq 2$ [9], $d_{n-1}(S_{n,k}) \geq d(S_{n,k}) + 1$. The lower bounds of $d_{n-1}(S_{n,k})$, where $2 \leq k \leq \lfloor n/2 \rfloor$, $d_{n-1}(S_{6,4})$, and $d_{n-1}(S_{5,3})$ are derived in Lemmas 3.9 to 3.11, respectively.

Lemma 3.9 [26]. $d_{n-1}(S_{n,k}) \geq d(S_{n,k}) + 2$, where $2 \leq k \leq \lfloor n/2 \rfloor$.

Lemma 3.10 $d_5(S_{6,4}) \geq d(S_{6,4}) + 2$.

Proof. Consider $\rho = 2143$ in $S_{6,4}$. Construct paths π_1 from 2143 via its neighbor 5143 to ε and π_2 from 2143 via its neighbor 6143 to ε such that their length are at most $d(S_{6,4}) + 1 = 7$. Figure 3 shows that π_1 and π_2 both contain vertex 2134 and they are not vertex-disjoint.

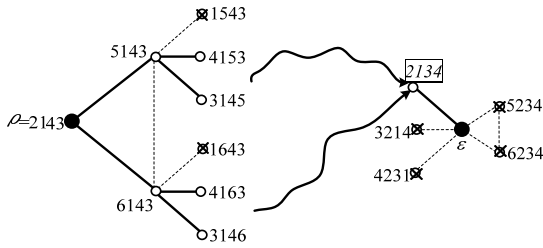


Figure 3. Two paths from 2143 to ε in an $S_{6,4}$.

By observation on the neighbors of 5143, vertex 1543 cannot be contained in π_1 , due to $d(S_{6,4}; 1543, 1234) = d(S_{6,4}) = 6$ and $l(\pi_1) = d(S_{6,4}) + 2 = 8$. Additionally, only one neighbor of ε , which is vertex 2134, can be considered for constructing a path of length less than $d(S_{6,4}) + 2$ because $d(S_{6,4}; 4153, 2134)$ and $d(S_{6,4}; 3145, 2134)$ are both less than 5. The case of constructing π_2 is similar to π_1 . Therefore, to exclude the common vertex 2134, the length of π_1 or π_2 should be at least $d(S_{n,k}) + 2$. $\square$

Lemma 3.11 $d_4(S_{5,3}) \geq d(S_{5,3}) + 2$.

Proof. Consider $\rho = 241$ in $S_{5,3}$. Construct paths π_1 from 241 via 541 to ε and π_2 from 241 via 523 to ε such that their length are at most $d(S_{5,3}) + 1$. Figure 4 shows that π_1 and π_2 both contain vertex 521 and they are not vertex-disjoint.

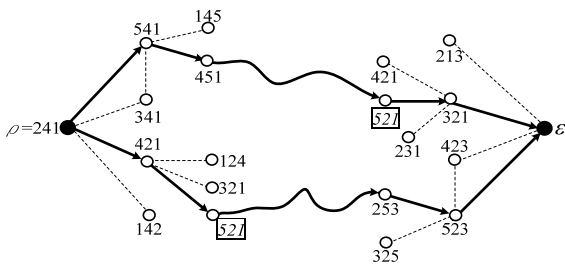


Figure 4. Two paths from 241 to ε in an $S_{5,3}$.

Because $d(S_{5,3}; 145, 123) = d(S_{5,3}) = 5$ and $l(\pi_1) = d(S_{5,3}) + 2 = 7$, only 451 can be included in π_1 . Furthermore, only $d(S_{5,3}; 451, 321) < 4$ and $d(S_{5,3}; 451, 521) < 3$, π_1 should be $241 \rightarrow 541 \rightarrow 451 \rightarrow \dots \rightarrow 521 \rightarrow 321 \rightarrow \varepsilon$. Similarly, π_2 should be $241 \rightarrow 421 \rightarrow 521 \rightarrow \dots \rightarrow 523 \rightarrow 523 \rightarrow \varepsilon$. Clearly, they both contain vertex 521. Therefore, to exclude the common vertex 521, the length of π_1 or π_2 should be at least $d(S_{n,k}) + 2$. $\square$

Lemmas 3.7 to 3.11 show that upper and lower bounds of $d_{n-1}(S_{n,k})$ are the same when $2 \leq k \leq \lfloor n/2 \rfloor$, or $\lfloor n/2 \rfloor + 1 \leq k \leq n - 2$ and $n < 7$. Therefore, a lemma is obtained as follows.

Lemma 3.12 $d_{n-1}(S_{n,k}) = d(S_{n,k}) + 2$ if $2 \leq k \leq \lfloor n/2 \rfloor$, or $\lfloor n/2 \rfloor + 1 \leq k \leq n - 2$ and $n < 7$.

When $\lfloor n/2 \rfloor + 1 \leq k \leq n - 2$ and $n \geq 7$, the lower and upper bounds of $d_{n-1}(S_{n,k})$ are not the same, and they are $d(S_{n,k}) + 1$ and $d(S_{n,k}) + 2$, respectively. Subsection 3.2.2 try to shorten the some constructed $C_{n-1}(S_{n,k}; \rho, \varepsilon)$ s by rebuild some paths of them. Consequently, the results demonstrate that lower bound and upper bound of $d_{n-1}(S_{n,k})$ are also the same and they are both $d(S_{n,k}) + 1$ when $\lfloor n/2 \rfloor + 1 \leq k \leq n - 2$ and $n \geq 7$.

3.2.2 Refinements for Some Containers

This subsection proposes four refinements for the constructed $C_{n-1}(S_{n,k}; \rho, \varepsilon)$ s with $\lfloor n/2 \rfloor + 1 \leq k \leq n - 2$ and $n \geq 7$. One of these refinements is adopted to shorten the path(s) in a $C_{n-1}(S_{n,k}; \rho, \varepsilon)$ which length is $d(S_{n,k}) + 2$. The longest path(s) in a $C_{n-1}(S_{n,k}; \rho, \varepsilon)$ can be represented by $d(S_{n,k}; \rho, \varepsilon)$ plus a non-negative integer, denoted by δ . Additionally, $d(S_{n,k}; \rho, \varepsilon) = d(S_{n,k}) - \Delta$, the longest path can thus be represented by $d(S_{n,k}) - \Delta_{min} + \delta$, where Δ_{min} denotes the specific vertex ρ with minimum Δ .

Moreover, according to Lemmas 2.3, 2.4 and Corollary 2.5, the parameters m, m', f and e' can be deduced by Δ_{min} . Hence, referring to the parameters m, m', f and e' , the cycle representation of ρ can be stated by a general form. The following work is to find the general form of a specific vertices ρ when $l(C_{n-1}(S_{n,k}; \rho, \varepsilon)) = d(S_{n,k}) + 2$, and to refine the longest path(s) in $C_{n-1}(S_{n,k}; \rho, \varepsilon)$ such that $l(C_{n-1}(S_{n,k}; \rho, \varepsilon))$ can be reduced at least by 1. The $C_{n-1}(S_{n,k}; \rho, \varepsilon)$ s of length $d(S_{n,k}) + 2$ are excerpted from Lemmas 3.2, 3.4 and 3.6, and Δ_{min} is listed in Table 1.

Assuming $\rho = p_1 p_2 \dots p_k$, $m = (k - 1) - f - e' = 2k - n - f + e' - 1$ ($m = k - f - e' = 2k - n - f + e'$) if $p_1 = 1$ ($p_1 \neq 1$) such that the number of internal cycle can be computed as $(m - m')/2$. Moreover, the number of inner external symbol also can be computed as $e - n - k - e'$. The cycle representation of a specific vertex ρ is thus stated in Table 1, Lemmas 3.13 to 3.16. For convenience, let m_i, d_i and e_i denote i th misplaced, desired, inner external symbols of the label of vertex ρ , respectively.

Table 1. The Δ_{min} , When $l(C_{n-1}(S_{n,k}; \rho, \varepsilon))=d(S_{n,k})+2$, and $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$, $n \geq 7$

Item	p_1	Path	Number of paths	Δ_{min}	Situations
1	$p_1=1$	γ	$n-k-e$	$\Delta_2=0$	n is even.
2	$p_1 \neq 1$	α	$m+2e-c-1$	$\Delta_2=0$	n is even.
3		β	$k-m-e$	$\Delta_2, \Delta_3=2$	
4		γ	$n-k-e$	$\Delta_2=1$	Symbol 1 is not a desired symbol, p_1 is external and n is even.
5				$\Delta_2, \Delta_3=1$	Symbol 1 is not a desired symbol, C_1 is internal and $e>0$.
6				$\Delta_2=2$	Symbol 1 is not a desired symbol, n is even and $e=0$.
7				$\Delta_2, \Delta_3=2$	Symbol 1 is not a desired symbol and $e>0$.
8				$\Delta_2=1$	Symbol 1 is a desired symbol, p_1 is internal and n is even.

Lemma 3.13 Assume that $p_1=1$, $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$, n is even and $n \geq 8$. If the length of a γ -path in a constructed $C_{n-1}(S_{n,k}; \rho, \varepsilon)$ is $d(S_{n,k})+2$, then ρ consists of at least 2 cycles.

Proof. Item 1 of Table 1 indicates that $\Delta_2=0$ and $e' \geq 1$ since there is at least one γ -path. By Lemma 2.3, $\Delta_2=(m'+e'+3f-1)/2=0$ and by Corollary 2.5, $m'+e'+3f=1$. Thus, $f=m'=0$ and $e'=1$ since $e' \geq 1$. Thus, $e=n-k-e'>0$ and $m=2k-n-f+e'-1=2k-n$. This implies that $c=(m-m')/2+1=(2k-n)/2+1 \geq 2$. Therefore, ρ consists of at least 2 cycles. $\square$

Lemma 3.14 Assume that $p_1 \neq 1$, $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$, n is even and $n \geq 8$. If the length of a α -path in a constructed $C_{n-1}(S_{n,k}; \rho, \varepsilon)$ is $d(S_{n,k})+2$, then ρ consists of at least 2 internal cycles and 1 external cycle which contains $n-k$ inner external symbols. Moreover, each internal cycle contains exactly 2 symbols and the external cycle has no redundant symbols.

Proof. Item 2 of Table 1 indicates that $\Delta_2=0$. By Lemma 2.3, $\Delta_2=(m'+e'+3f)/2=0$ and by Corollary 2.5, $m'+e'+3f=0$. Therefore, $m'=e'=f=0$ and this implies that each internal cycle contains exactly 2 symbols. Since $e'=0$ and $m=2k-n-f+e'=2k-n$, ρ consists of $n-k$ inner external symbols. Therefore, ρ consists of $(m-m')/2=k-n/2 \geq 2$ internal cycles and 1 external cycle. $\square$

Lemma 3.15 Assume that $p_1 \neq 1$, $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n \geq 7$. If the length of a β -path in a constructed $C_{n-1}(\rho, \varepsilon)$ is $d(S_{n,k})+2$, then ρ consists of at least 2 cycles, or ρ only consists of one external cycle which contains at least 3 inner external symbols.

Proof. Item 3 of Table 1 indicates that $\Delta_2=2$ or $\Delta_3=2$ and $f \geq 1$ since there is at least a β -path. By Lemma 2.3, $\Delta_2=(m'+e'+3f)/2=2$ and $\Delta_3=(m'+e'+3f+1)/2=2$, and by Corollary 2.5, $m'+e'+3f=4$ ($m'+e'+3f+1=4$) if n is even (odd). Therefore, $f=1$, $m'+e'=1$ ($m'+e'=0$) if n is even (odd) and $e=n-k-e'>0$. Since $m=2k-n-f+e'=2k-n+e'-1$, $c=(m-m')/2+1=(2k-n+e'-1-m')/2+1$. Two cases are needed to consider as follows.

Case 1 (n is even).

Because $m'+e'=1$, $m'=0$ and $e'=1$, or $m'=1$ and $e'=0$. If $m'=0$ and $e'=1$ then $c=(2k-n)/2+1 \geq 2$. If $m'=1$ and $e'=0$ then $c=(2k-n-2)/2+1$. Consequently, $c=1$ when

$k=\lfloor n/2 \rfloor + 1$, or $c \geq 2$ when $k > \lfloor n/2 \rfloor + 1$. The number of cycles $c=1$ implies that $e=n-k-e'=n-\lfloor n/2 \rfloor -1 \geq 3$ and ρ only consists only 1 external cycle which contains at least 3 inner external symbols. The number of cycles $c \geq 2$ implies that ρ consists of at least 2 cycles.

Case 2 (n is odd).

Because $m'+e'=0$, $m'=e'=0$, and $c=(2k-n-1)/2+1$. Thus, $c=1$ when $k=\lfloor n/2 \rfloor + 1$, or $c \geq 2$ when $k > \lfloor n/2 \rfloor + 1$. The number of cycles $c=1$ implies that $e=n-k-e'=n-\lfloor n/2 \rfloor -1 \geq 3$ and ρ only consists only 1 external cycle which contains at least 3 inner external symbols. The number of cycles $c \geq 2$ implies that ρ consists of at least 2 cycles. $\square$

Lemma 3.16 Assume that $p_1 \neq 1$, $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n \geq 7$. If the length of a γ -path in a constructed $C_{n-1}(\rho, \varepsilon)$ is $d(S_{n,k})+2$, then ρ can be one of the four general forms:

If $e'=1$,

Form 1. At least 2 cycles.

Form 2. Only external cycle contains at least 2 inner external symbols.

If $e'=2$,

Form 3. At least 3 cycles.

Form 4. At least 2 cycles, and the C_1 is formed as $(m_1 e_1 \dots d_1 1)$.

Proof. Table 1 shows that the length of a γ -path is $d(S_{n,k})+2$ and should be one of following situations.

(1) According to Items 4, 5, 6 and 7, symbol 1 is not a desired symbol:

(1.1) According to Item 4, p_1 is external, n is even and $\Delta_2=1$: C_1 can be formed as $(e_1 \dots d_1 1)$, where $p_1=e_1$ and $d_1 \neq 1$ such that $m' \geq 1$. Given $e' \geq 1$, by Lemma 2.3, $\Delta_2=(m'+e'+3f)/2=1$ and by Corollary 2.5, $m'+e'+3f=2$. Thus, $f=0$, $m'=e'=1$ since $e', m' \geq 1$, and $e>0$. Moreover, since $m=2k-n-f+e'=2k-n+1 \geq 3$, $c=(m-m')/2+1 \geq 2$. This situation belongs to Form 1.

(1.2) According to Item 5, C_1 is internal, $e>0$ and $\Delta_2=\Delta_3=1$: Given $e' \geq 1$, by Lemma 2.3, $\Delta_2=(m'+e'+3f)/2=1$ and $\Delta_3=(m'+e'+3f+1)/2=1$, and by Corollary 2.5, $m'+e'+3f=2$ ($m'+e'+3f=1$) if n is even (odd). Thus, $f=0$ and $m'+e'=2$ ($=1$). Two subcases are considered as follows.

- (1.2.1) n is even: If $m'=e'=1$, then $m=2k-n-f+e'=2k-n+1 \geq 3$. Hence, $c=(m-m')/2+1 \geq 2$. This situation belongs to Form 1. If $m'=0$ and $e'=2$, then $m \geq 4$. Hence, $c \geq 3$. This situation belongs to Form 3.
- (1.2.2) n is odd: If $m'=0$ and $e'=1$, then $m \geq 2$ and $c \geq 2$. This situation belongs to Form 1.
- (1.3) According to Item 6, n is even and $e=0$. Hence, $e' \geq 2$. Given $\Delta_2=2$, by Lemma 2.4, $\Delta_2=(m'+e'+3f)/2+1=2$, and by Corollary 2.5, $m'+e'+3f=2$. Thus, $f=0$, $m'=0$ and $e'=2$. Moreover, $e=0$ and $e'=2$ imply $k=n-2$. Therefore, $m=k-f-e=k=n-2 \geq 6$ and $c=(m-m')/2 \geq 3$. This situation belongs to Form 3.
- (1.4) According to Item 7, C_1 is an external cycle and p_1 is internal. Since $e>0$, C_1 contains at least four symbols and can be formed as $(m_1 e_1 \dots d_1 1)$. Given $\Delta_2=\Delta_3=2$, by Lemma 2.3, $\Delta_2=(m'+e'+3f)/2=2$ and $\Delta_3=(m'+e'+3f+1)/2=2$, and by Corollary 2.5, $m'+e'+3f=4$ ($m'+e'+3f=3$) if n is even (odd). Thus, $f=0$ and $m'+e'=4$ ($=3$). Two subcases are considered as follows.
- (1.4.1) n is even: If $m'=e'=2$, then $m=2k-n-f+e'=2k-n+2 \geq 4$. Hence, $c=(m-m')/2+1 \geq 2$. This situation belongs to Form 4. If $m'=3$, $e'=1$, and $k=\lfloor n/2 \rfloor + 1$, then $m=2k-n-f+e'=2k-n+1=3$ and $e=n-k-e' \geq 2$. Hence, $c=(m-m')/2+1 \geq 1$. This situation belongs to Form 2. If $m'=3$, $e'=1$, and $k > \lfloor n/2 \rfloor + 1$, then $m=2k-n+1 \geq 5$. Hence $c=(m-m')/2+1 \geq 2$. This situation belongs to Form 1.
- (1.4.2) n is odd: If $m'=2$ and $e'=1$, and $k=\lfloor n/2 \rfloor + 1$, then $m=2k-n+1=2$ and $e=n-k-e' \geq 2$. Hence, $c=(m-m')/2+1 \geq 1$. This situation belongs to Form 2. If $m'=2$ and $e'=1$, and $k > \lfloor n/2 \rfloor + 1$, then $m \geq 4$. Hence, $c=(m-m')/2+1 \geq 2$. This situation belongs to Form 1.
- (2) According to Item 8, symbol 1 is a desired symbol, p_1 is internal and n is even: C_1 can be formed as $(m_1 e_1 1)$ and $m' \geq 1$. Given $\Delta_2=1$, by Lemma 2.3, $\Delta_2=(m'+e'+3f)/2=1$, and by Corollary 2.5, $m'+e'+3f=2$. Thus, $f=0$ and $m'+e'=2$. Additionally, $m'=e'=1$ because $e' \geq 1$ and $m' \geq 1$. Since $m=2k-n-f+e'=2k-n+1 \geq 3$, $c=(m-m')/2+1 \geq 2$. This situation belongs to Form 1. $\square$

According to Table 1, Lemmas 3.13 to 3.16, the general form of the vertex ρ is stated in Table 2. Based on this table, the refinements are proposed in the following. Significantly, at most one of the four refinements should be adopted.

3.2.2.1 Refinement 1: α -Paths Reconstruction

By Lemma 3.14, $e'=f=0$ implies no β -path and γ -path to be constructed. Thus, this refinement is proposed to reconstruct all of $n-1$ vertex-disjoint α -paths for vertex ρ .

Assuming $\rho=C_1C_2\dots C_x$, $x \geq 4$, where the external cycle is separated into $n-k \geq 2$ individual external cycles. Let $C_i=(s_{i,1} s_{i,2})$ for each i , $1 \leq i \leq x$, since no redundant symbol exists in each cycle. The refinement is to divide cycles $C_2, \dots, C_x$ into two parts C_i and $C_{i+1}\dots C_xC_2\dots C_{i-1}$, for each i , $2 \leq i \leq x$. The $n-1$ α -paths are reconstructed in 4 Groups:

Group 1. $x-1$ paths $\pi(s_{i,1})$ are constructed by correcting $C_i C_1$ and then $C_{i+1}\dots C_xC_2\dots C_{i-1}$, for each i , $2 \leq i \leq x$. The sequence $\sigma(s_{i,1})$ is arranged as $\langle s_{i,1}, s_{i,2}, C_1, s_{i-1,2}, C_{i+1}, \dots, C_x, C_2, \dots, C_{i-2}, s_{i-1,1} \rangle$, or $\langle s_{i,1}, s_{i,2}, C_1, s_{x,2}, C_{i+1}, \dots, C_x, C_2, \dots, C_{i-2}, s_{x,1} \rangle$, if $i=2$. Notably, $C_{i+1}\dots C_xC_2\dots C_{i-2}$ are between $s_{i-1,2}$ and $s_{i-1,1}$, and the correcting order of $s_{i,1}$ and $s_{i-1,1}$ is the first and last symbol of the sequence.

Group 2. $x-2$ paths $\pi(s_{j+1,2})$ are constructed by correcting $C_{j+1}\dots C_xC_2\dots C_{j-1}C_1$ and then C_j , for each j , $2 \leq j \leq x-1$. The sequence $\sigma(s_{j+1,2})$ is arranged as $\langle s_{j+1,2}, C_{j+2}, \dots, C_x, C_2, \dots, C_{j-1}, s_{j+1,1}, C_1, s_{j,1}, s_{j,2} \rangle$. Notably, $C_{j+2}\dots C_xC_2\dots C_{j-1}$ are between $s_{j+1,2}$ and $s_{j+1,1}$, and the correcting order of $s_{j+1,2}$ and $s_{j,2}$ is the first and last symbol of the sequence.

Group 3. Path $\pi(s_{1,1})$ is constructed by $\sigma(s_{1,1})=\langle C_1, s_{x,1}, C_2, \dots, C_{x-1}, s_{x,2} \rangle$.

Group 4. Path $\pi(s_{2,2})$ is constructed by $\sigma(s_{2,2})=\langle s_{2,2}, C_3, \dots, C_x, s_{2,1}, C_1 \rangle$.

In total, $2x-1=2(n/2)-1=n-1$ α -paths are reconstructed. Lemma 3.17 shows that the length of each constructed path is at most $d(S_{n,k})+1$.

Lemma 3.17 There are $n-1$ vertex-disjoint α -paths of length at most $d(S_{n,k})+1$ reconstructed by Refinement 1.

Proof. The paths constructed by Group 1 are vertex-disjoint, because the correcting order of $s_{i-1,1}$ is prior to $s_{i,1}$ in each $\sigma(s_{i,1})$, $2 \leq i \leq x$. Therefore, by Lemma 2.1, $\pi(s_{j,1})$ and $\pi(s_{i,1})$ are vertex-disjoint. Similarly, $\pi(s_{j,2})$ and $\pi(s_{i,2})$ are vertex-disjoint for each $\pi(s_{j,2})$ in Group 2. Let $\sigma(s_{i,1})$ and $\sigma(s_{j,2})$ be the sequences of Group 1 and Group 2, where $2 \leq i \leq x$ and $2 \leq j \leq x-1$. To show Group 1 and Group 2 are also vertex-disjoint to each other, two cases are needed to consider as follows.

Case 1. ($C_i \neq C_{j+1}$).

Clearly, C_{i-1} is ahead of C_i in the $\sigma(s_{j,2})$ and the correcting order of $s_{i-1,1}$ is prior to $s_{i,1}$. By Lemma 2.1, $\pi(s_{j,2})$ and $\pi(s_{i,1})$ are vertex-disjoint to each other.

Table 2. The Characteristics of Vertex ρ , When $l(C_{n-1}(S_{n,k}; \rho, \varepsilon)) = d(S_{n,k}) + 2$, and $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$, $n \geq 7$

Path	p_1	The Characteristics of Vertex ρ	Δ_2	Δ_3
α	$p_1 \neq 1$	Vertex ρ contains at least 2 internal cycles and the external cycle contains $n-k$ inner external symbols. No redundant symbol exists in each cycle.	0	NA
$\beta (f=1)$	$p_1 \neq 1$	Vertex ρ contains at least 2 cycles or only the external cycle which has at least 3 external symbols.	2	2
$\gamma (e'=1)$	$p_1 = 1$	Vertex ρ contains at least 2 cycles.	0	NA
	$p_1 \neq 1$	Vertex ρ contains at least 2 cycles.	1 or 2	1 or 2
		Vertex ρ only the external cycle which has at least 2 external symbols.	2	2
$\gamma (e'=2)$	$p_1 \neq 1$	Vertex ρ contains at least 3 cycles.	1 or 2	1 or 2
		At least 2 cycles and one of them is the external cycle $C_1 = (m_1 e_1 \dots d_1 1)$.	2	2

Case 2. ($C_i = C_{j+1}$).

The sequence $\sigma(s_{j,2})$ can be arranged as $\langle s_{i,2}, C_{i+1}, \dots, C_x, C_2, \dots, C_{i-2}, s_{i,1}, C_1, s_{i-1,1}, s_{i-1,2} \rangle$ and $\sigma(s_{i,1})$ as $\langle s_{i,1}, s_{i,2}, C_1, s_{i-1,2}, C_{i+1}, \dots, C_x, C_2, \dots, C_{i-2}, s_{i-1,1} \rangle$. Divides sequence $\sigma(s_{j,2})$ into two subsequences $\sigma(s_{j,2})_1 = \langle s_{i,2}, C_{i+1}, \dots, C_x, C_2, \dots, C_{i-2}, s_{i,1} \rangle$ and $\sigma(s_{j,2})_2 = \langle C_1, s_{i-1,1}, s_{i-1,2} \rangle$ and sequence $\sigma(s_{i,1})$ into two subsequences $\sigma(s_{i,1})_1 = \langle s_{i,1}, s_{i,2}, C_1 \rangle$ and $\sigma(s_{i,1})_2 = \langle s_{i-1,2}, C_{i+1}, \dots, C_x, C_2, \dots, C_{i-2}, s_{i-1,1} \rangle$. The paths $\pi(s_{j,2})$ ($\pi(s_{i,1})$) can thus be constructed by $\sigma(s_{j,2})_1$ and $\sigma(s_{j,2})_2$ ($\sigma(s_{i,1})_1$ and $\sigma(s_{i,1})_2$). Two subpaths constructed by $\sigma(s_{j,2})_1$ and $\sigma(s_{j,2})_2$ ($\sigma(s_{i,1})_1$ and $\sigma(s_{i,1})_2$) are denoted by $\pi(s_{j,2})_1$ and $\pi(s_{j,2})_2$ ($\pi(s_{i,1})_1$ and $\pi(s_{i,1})_2$), respectively. By Corollary 2.2, the subpaths $\pi(s_{j,2})_1$ and $\pi(s_{i,1})_1$ are vertex-disjoint, because $s_{i,1}$ is prior to $s_{i,2}$ in $\sigma(s_{i,1})_1$. The subpaths $\pi(s_{j,2})_1$ and $\pi(s_{i,1})_2$ are vertex-disjoint, because the cycle C_1 has already been corrected in $\pi(s_{i,1})_2$, but $\pi(s_{j,2})_1$ has not yet been corrected. Because $x \geq 4$, there exists a cycle C_{i+1} in ρ . The subpaths $\pi(s_{j,2})_2$ and $\pi(s_{i,1})_1$ are vertex-disjoint, because the cycle C_{i+1} has already been corrected in $\pi(s_{j,2})_2$, but $\pi(s_{i,1})_1$ has not yet been corrected. By Corollary 2.2, the subpaths $\pi(s_{j,2})_2$ and $\pi(s_{i,1})_2$ are vertex-disjoint, because $s_{i-1,1}$ is prior to $s_{i-1,2}$ in $\pi(s_{j,2})_2$. Consequently, $\pi(s_{i,1})$ and $\pi(s_{j,2})$ are vertex-disjoint.

By Lemma 2.1, the path $\pi(s_{1,1})$ in Group 3 is vertex-disjoint to the paths in Groups 1 because $s_{i-1,1}$ is prior to $s_{i,1}$ in $\sigma(s_{1,1})$, and is vertex-disjoint to the paths in Groups 2 because $s_{x,2}$ is prior to $s_{1,1}$ in $\sigma(s_{j,2})$. The path $\pi(s_{2,2})$ in Group 4 is vertex-disjoint to the paths in Groups 1 because $s_{1,2}$ is prior to $s_{2,2}$ in $\sigma(s_{i,1})$ if $i \neq 2$, and $s_{x,1}$ is prior to $s_{2,1}$ in the $\sigma(s_{2,2})$ if $i=2$. The path $\pi(s_{2,2})$ in Group 4 is vertex-disjoint to the paths in Groups 2 because $s_{j,2}$ is prior to $s_{j+1,2}$ in $\sigma(s_{2,2})$, and is vertex-disjoint to the paths in Groups 3 because $s_{x,2}$ is prior to $s_{1,1}$ in the $\sigma(s_{2,2})$. Therefore, the $n-1$ paths are vertex-disjoint.

It is given that ρ has $c-1$ internal cycles and $n-k$ external symbols in the external cycle. Let the cycle representation of ρ is $C_1 C_2 \dots C_c$. Two cases are considered as follows: Case a. If C_1 is an internal cycle then without loss of generality, let C_c be the external cycle and separated into individual external cycles $C'_c \dots C'_x$, such that $C_1 C_2 \dots C_c$ become $C_1 C_2 \dots C_{c-1} C'_c \dots C'_x$. Case b. If C_1 is an external cycle then separate C_1 into individual external cycles

$C'_1 C'_2 \dots C'_{n-k}$ and renumber $C_2 \dots C_c$ as $C_{n-k+1} \dots C_x$, such that $C_1 C_2 \dots C_c$ become $C'_1 C'_2 \dots C'_{n-k} C_{n-k+1} \dots C_x$. The length of each path is computed as follows:

Case a. (C_1 is an internal cycle).

The length of each path in Groups 1 and 2 is at most $d(S_{n,k}; \rho, \varepsilon) + 1$, since in worst case some external cycle is prior to C_1 and some external cycle is posterior to C_1 such that the number of cycle is thus increased by 1. The length of each path in Groups 3 and 4 is at most $d(S_{n,k}; \rho, \varepsilon)$, since it is a shortest path. Notably, both Groups 3 and 4 have single path. Referring to the 1st row of Table 2, $\Delta_2 = 0$ and the length of each reconstructed path is thus at most $d(S_{n,k}) + 1$.

Case b. (C_1 is an external cycle).

The external cycle C_1 is divided into individual external cycles; in Groups 1 and 2, in worst case some external cycle is prior to a symbol and some external cycle is posterior to the symbol. Thus, the length of each path in Groups 1 and 2 is at most $d(S_{n,k}; \rho, \varepsilon) + 1$, since the number of cycle is increased by 1. Therefore, the length of each reconstructed path is at most $d(S_{n,k}) + 1$. $\square$

In one-to-one vertex-disjoint algorithm, the construction of the β -paths and γ -paths has an extra step to move one fixed symbol β_i or outer external symbol γ_i to position 1 and never correct it until all other symbols have been corrected. Thus, two additional edges are added. Moreover, in order to reserve the correction of β_i or γ_i last, one (or two) additional edge(s) would be further added. In total, the length of the path is increased to $d(S_{n,k}; \rho, \varepsilon) + 3$ (or +4) if $p_1 \neq 1$, or $d(S_{n,k}; \rho, \varepsilon) + 2$ if $p_1 = 1$. Therefore, $d_{n-1}(S_{n,k}; \rho, \varepsilon)$ is at most $d(S_{n,k}) + 2$. Our goal is to shorten the path(s) of length $d(S_{n,k}) + 2$. To achieve the goal, β_i and γ_i cannot be corrected last. In order to reconstruct the longest paths, some functions used in Refinements 2-4 are first defined as follows.

Assuming a, b are two distinct symbols, and x_i is a symbol or cycle of a vertex. Two functions shown below are defined for reconstructing the β -path and γ -path(s) whose lengths are $d(S_{n,k}) + 2$.

1. **$X(a, b)$:** When symbol a is at position 1, replace a by b or exchange a and b .

2. $F(x_1, x_2, \dots, x_t)$: Correct symbols or cycles in the order of $x_1, x_2, \dots, x_t$.

3.2.2.2 Refinement 2: β -Path Reconstruction

This refinement is for the β -path as shown in the 2nd row of Table 2. Lemma 3.15 indicates that only one $\pi(\beta_i)$ should be refined since $f=1$. Two cases of the refinement are stated below.

Case 1. (ρ has at least 2 cycles).

Notably, only one γ -path is constructed in this case, since only one outer external symbols exists in ρ . Let ρ be $C_1 C_2 \dots C_c$, where $c \geq 2$. Path $\pi(\beta_1)$ can be constructed by performing $X(p_1, \beta_1)$ and $F(C_2, \dots, C_c, \beta_1, C_1)$. In total, one edge is contributed by $X(p_1, \beta_1)$; $c-1$ heads apply rule R2, and $m+2e-(c-1)$ symbols apply rule R1. Referring to the 2nd row of Table 2, $\Delta_2=\Delta_3=2$ and the length of the reconstructed $\pi(\beta_1)$ is thus $1+2(c-1)+(m+2e-c+1) = c+m+2e = d(S_{n,k}; \rho, \varepsilon)+2 = d(S_{n,k})$.

However, the reconstructed $\pi(\beta_1)$ and the constructed α -path $\pi(s_{2,1})$ by the sequence $\langle C_2, \dots, C_c, C_1 \rangle$ intersect at vertex $U=C_1$. Hence, $\pi(s_{2,1})$ should be reconstructed to exclude vertex U by performing $X(p_1, s_{2,1})$ and $X(s_{2,1}, s_{1,k_1})$ such that $\langle C_2 s_{1,1} \dots s_{1,k_1-1} \rangle$ forms a new cycle, and then perform $F(s_{1,1}, \dots, s_{1,k_1-1}, C_2, C_3, \dots, C_c)$. In total, two edges are contributed by $X(p_1, s_{2,1})$ and $X(s_{2,1}, s_{1,k_1})$; $c-1$ heads apply rule R2, and $m+2e-1-(c-1)$ symbols apply rule R1. Thus, the length of the reconstructed $\pi(s_{2,1})$ is $2+2(c-1)+(m+2e-c) = c+m+2e = d(S_{n,k}; \rho, \varepsilon)+2 = d(S_{n,k})$, since $\Delta_2=\Delta_3=2$.

Unfortunately, the reconstructed $\pi(s_{2,1})$ and the constructed α -path $\pi(s_{1,1})$ by the sequence $\langle C_1, \dots, C_c \rangle$ intersect at vertex $V=\langle C_2 \rangle C_3 \dots C_c$. Notably, cycle $\langle C_2 \rangle$ indicates $X(1, s_{2,1})$ has been performed. Hence, $\pi(s_{1,1})$ should be reconstructed to exclude vertex V by performing $F(C_1)$ and $X(1, \beta_1)$ and then $F(C_2, \dots, C_c, \beta_1)$. In total, one edge is contributed by $X(1, \beta_1)$; $c-1$ heads apply rule R2, and $m+2e-(c-1)$ symbols apply rule R1. Thus, the length of the reconstructed $\pi(s_{1,1})$ is $1+2(c-1)+(m+2e-c+1) = c+m+2e = d(S_{n,k}; \rho, \varepsilon)+2 = d(S_{n,k})$, since $\Delta_2=\Delta_3=2$. Figure 5 illustrates the reconstructed paths.

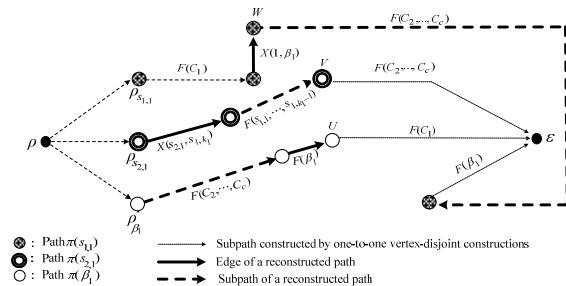


Figure 5. Case 1 of Refinement 2.

Let $[u-v]_\pi$ denote the subpath of π from u to v including u and v . Moreover, adopt "(" or ")" for

excluding u (left parentheses) or v (right parentheses). To show the refinement can reconstruct a new $C_{n-1}(S_{n,k}; \rho, \varepsilon)$, according to Figure 5, the three subpaths $(\rho_{\beta_1}-U)_{\pi(\beta_1)}$, $[W-\varepsilon]_{\pi(s_{1,1})}$ and $(\rho_{s_{2,1}}-V)_{\pi(s_{2,1})}$ should be verified that they are vertex-disjoint to each other and vertex-disjoint to all α -paths and the γ -path as follows.

1. Symbol β_1 is not fixed only in each vertex of subpaths $(\rho_{\beta_1}-U)_{\pi(\beta_1)}$ and $[W-\varepsilon]_{\pi(s_{1,1})}$. Therefore, the reconstructed subpaths $(\rho_{\beta_1}-U)_{\pi(\beta_1)}$ and $[W-\varepsilon]_{\pi(s_{1,1})}$ are vertex-disjoint to the γ -path, all α -paths, and $(\rho_{s_{2,1}}-V)_{\pi(s_{2,1})}$.
2. Subpath $(\rho_{s_{2,1}}-V)_{\pi(s_{2,1})}$ is obviously vertex-disjoint to the γ -path. After performing $X(s_{2,1}, s_{1,k_1})$ on $\rho_{s_{2,1}}$, symbols $(s_{2,1} \dots s_{2,k_2} s_{1,1} \dots s_{1,k_1-1})$ become a new cycle. Thus, symbol $s_{2,1}$ is at position s_{1,k_1-1} in each vertex of $(\rho_{s_{2,1}}-V)_{\pi(s_{2,1})}$, but each vertex in all α -paths has no symbol $s_{2,1}$ residing at position s_{1,k_1-1} . Therefore, $(\rho_{s_{2,1}}-V)_{\pi(s_{2,1})}$ and other α -paths are vertex-disjoint. Consequently, the reconstructed $\pi(s_{2,1})$ is vertex-disjoint to other α -paths and the γ -path.
3. The subpaths $(\rho_{\beta_1}-U)_{\pi(\beta_1)}$ and $[W-\varepsilon]_{\pi(s_{1,1})}$ are vertex-disjoint, because C_1 has already been corrected in $[W-\varepsilon]_{\pi(s_{1,1})}$ but $(\rho_{\beta_1}-U)_{\pi(\beta_1)}$ has not yet.

Case 2. (The external cycle is the only cycle of ρ and $e \geq 3$).

Notably, no γ -paths are constructed in this case, since no outer external symbols exist in ρ . The path $\pi(\beta_1)$ is reconstructed by separating the external cycle into two parts.

Assuming $\rho=(C_{1,1}C_{1,2}\dots C_{1,e})=(s_{1,1} s_{1,2} \dots s_{1,k_1-1} s_{1,k_1})$, where $C_{1,i}$, $1 \leq i \leq e$, is i th individual external cycle of ρ . Note that $s_{1,1}$ and s_{1,k_1} belong to $C_{1,1}$. Now separate $(C_{1,1}C_{1,2}\dots C_{1,e})$ into C'_1 and C'_2 two parts such that C'_1 includes symbol s_{1,k_1-1} . Without loss of generality, assuming C'_2 is $(s_{1,x} \dots s_{1,y})$, where $x \geq 2$ and $y < k_1-1$, and C'_1 can then be formed as $(s_{1,1} \dots s_{1,x-1} s_{1,y+1} \dots s_{1,k_1-1} s_{1,k_1})$. Moreover, C'_1 and C'_2 can be still formed as external cycle. Path $\pi(\beta_1)$ is reconstructed by performing $X(p_1, \beta_1)$ and $F(C'_2, \beta_1, C'_1)$. In total, one edge is contributed by $X(p_1, \beta_1)$; the head $s_{1,x}$ of C'_2 applies rule R2, and $m+2e-1$ symbols apply rule R1. Thus, the length of the reconstructed $\pi(\beta_1)$ is $1+2+(m+2e-1) = m+2e+2 = c+m+2e+1 = d(S_{n,k}; \rho, \varepsilon)+3 = d(S_{n,k})+1$, where $c=1$, since $\Delta_2=\Delta_3=2$.

Notably, $\pi(\beta_1)$ and $\pi(s_{1,1})$ intersect at the vertex $U=(s_{1,y+1} \dots s_{1,k_1-1} s_{1,k_1})$. Hence, $\pi(s_{1,1})$ is reconstructed to avoid the vertex U by performing $F(C'_1)$ and $X(1, \beta_1)$ and then $F(C'_2, \beta_1)$. In total, one edge is contributed by performing $X(1, \beta_1)$; the head $s_{1,x}$ of C'_2 applies rule R2, and $m+2e-c$ symbols apply R1. The length of the reconstructed $\pi(s_{1,1})$ is $1+2+(m+2e-1) = c+m+2e+1 = d(S_{n,k}; \rho, \varepsilon)+3 = d(S_{n,k})+1$, where $c=1$, since $\Delta_2=\Delta_3=2$. Figure 6 illustrates the reconstructed paths.

According to Figure 6, two subpaths $(\rho_{s_{1,1}}-\varepsilon)_{\pi(s_{1,1})}$

and $(\rho_{\beta_1}-U)_{\pi(\beta_1)}$ should be verified that they are vertex-disjoint to each other and vertex-disjoint to all α -paths as follows.

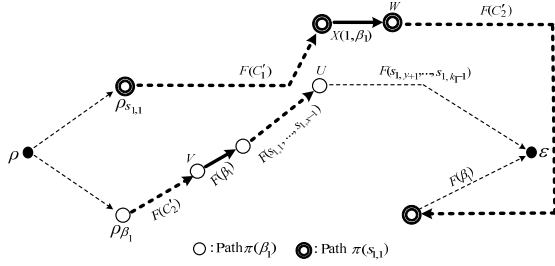


Figure 6. Case 2 of Refinement 2.

1. Symbol β_1 is not fixed only in each vertex of subpaths $(\rho_{\beta_1}-V)_{\pi(\beta_1)}$ and $[W-\varepsilon]_{\pi(s_{1,1})}$. Therefore, the reconstructed subpaths $(\rho_{\beta_1}-V)_{\pi(\beta_1)}$ and $[W-\varepsilon]_{\pi(s_{1,1})}$ are vertex-disjoint to all α -paths. Moreover, according Corollary 2.2, subpath $(V-U)_{\pi(\beta_1)}$ is vertex-disjoint to all α -paths, because $s_{1,1}$ is prior to $s_{1,x-1}$ in the sequence of $(V-U)_{\pi(\beta_1)}$, and subpath $(\rho_{s_{1,1}}-W)_{\pi(s_{1,1})}$ is vertex-disjoint to other α -paths, because $s_{1,1}$ is prior to the symbols of C'_2 in the sequence of $(\rho_{s_{1,1}}-W)_{\pi(s_{1,1})}$. Therefore, the reconstructed subpaths $(\rho_{s_{1,1}}-\varepsilon)_{\pi(s_{1,1})}$ and $(\rho_{\beta_1}-U)_{\pi(\beta_1)}$ are vertex-disjoint to all α -paths other than the original $\pi(s_{1,1})$.
2. Subpaths $(\rho_{\beta_1}-V)_{\pi(\beta_1)}$ and $[W-\varepsilon]_{\pi(s_{1,1})}$ are vertex-disjoint, because $s_{1,1}$ has already been corrected in $[W-\varepsilon]_{\pi(s_{1,1})}$ but $(\rho_{\beta_1}-V)_{\pi(\beta_1)}$ has not yet. Subpaths $(\rho_{\beta_1}-V)_{\pi(\beta_1)}$ and $(\rho_{s_{1,1}}-W)_{\pi(s_{1,1})}$ are vertex-disjoint, because β_1 is fixed in $(\rho_{s_{1,1}}-W)_{\pi(s_{1,1})}$ but $(\rho_{\beta_1}-V)_{\pi(\beta_1)}$ is not. Subpaths $(V-U)_{\pi(\beta_1)}$ and $(\rho_{s_{1,1}}-W)_{\pi(s_{1,1})}$ are vertex-disjoint, because C'_2 has already been corrected in $(V-U)_{\pi(\beta_1)}$ but $(\rho_{s_{1,1}}-W)_{\pi(s_{1,1})}$ has not yet. Subpaths $(V-U)_{\pi(\beta_1)}$ and $[W-\varepsilon]_{\pi(s_{1,1})}$ are vertex-disjoint, because β_1 is fixed in $(V-U)_{\pi(\beta_1)}$ but $[W-\varepsilon]_{\pi(s_{1,1})}$ is not. Therefore, the reconstructed subpaths $(\rho_{s_{1,1}}-\varepsilon)_{\pi(s_{1,1})}$ and $(\rho_{\beta_1}-U)_{\pi(\beta_1)}$ are vertex-disjoint to each other.

Based on the discussion above, the following lemma can be obtained.

Lemma 3.18 There are $n-1$ vertex-disjoint paths from ρ to ε of length at most $d(S_{n,k})+1$ after reconstruction of Refinement 2.

3.2.2.3 Refinement 3: γ -Path Reconstruction When $e'=1$

This refinement is designed for a γ -path when $e'=1$ as shown in the 3rd row of Table 2. Notably, no β -paths are constructed in this situation, since no fixed symbols exist in ρ . Two cases of the refinement for $p_1=1$ and $p_1 \neq 1$ are stated below.

Case 1. ($p_1=1$).

Notably, ρ has at least 2 cycles and $e=n-k-e' \geq 1$ implies ρ has an external cycle. Without loss of

generality, assuming $\rho=C_1C_2\dots C_c$, where $c \geq 2$ and $C_c=(d_1 \dots d_e e_e)$ is the external cycle. Because $\rho_{\gamma_1}=(\gamma_1 1)C_1C_2\dots C_c$, both $(\gamma_1 1)$ and C_c are external, they should be combined to form an external cycle $(\gamma_1 d_1 \dots d_e e_e 1)$ and denoted by C'_c . Restated, symbol 1 is also represented by s_{1,k_1} in this case. The cycle representation of ρ_{γ_1} is then $C'_cC_1\dots C_{c-1}$. Reconstruct the γ -path $\pi(\gamma_1)$ from ρ via ρ_{γ_1} to ε by performing $X(p_1, \gamma_1)$ and $F(C_1, \dots, C_{c-1}, C'_c)$. In total, one edge is contributed to connect ρ and ρ_{γ_1} ; $c-1$ heads apply R2, and $m+2e+1-(c-1)$ symbols apply R1. Referring to the 3th row of Table 2, $\Delta_2=0$ and the length of reconstructed $\pi(\gamma_1)$ is thus $1+2(c-1)+(m+2e-c+2) = c+m+2e+1 = d(S_{n,k}; \rho, \varepsilon)+1 = d(S_{n,k})+1$.

However, the reconstructed $\pi(\gamma_1)$ and the constructed α -path $\pi(s_{1,1})$ by the sequence $\langle C_1, \dots, C_c \rangle$ intersect at the vertex $U=C_c=(d_1 \dots d_e e_e 1)$ as shown in Figure 7. Hence, $\pi(s_{1,1})$ should be reconstructed. Path $\pi(s_{1,1})$ is reconstructed by performing $F(s_{1,1}, C_2, \dots, C_c, s_{1,2}, \dots, s_{1,k_1})$. In total, c heads apply R2 and $m+2e-c$ symbols apply R1. Hence, the length of the reconstructed $\pi(s_{1,1})$ is $c+m+2e = d(S_{n,k}; \rho, \varepsilon) = d(S_{n,k})$, since $\Delta_2=0$.

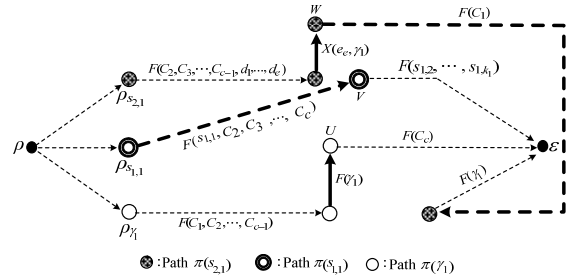


Figure 7. Case 1 of Refinement 3.

Unfortunately, the reconstructed paths $\pi(s_{1,1})$ and the α -path $\pi(s_{2,1})$ also intersect at vertex $V=(s_{1,2} \dots s_{1,k_1-1} s_{1,k_1})$ as shown in Figure 7, because $\pi(s_{2,1})$ is constructed by $\langle C_2, \dots, C_c, C_1 \rangle$, and when $C_2, \dots, C_c$ and $s_{1,1}$ are fixed, both paths contain vertex V . Hence, $\pi(s_{2,1})$ should be reconstructed by first performing $F(C_2, C_3, \dots, C_{c-1}, d_1, \dots, d_e)$, where $d_1, \dots, d_e$ are the symbols in C_c , to reach the vertex $(e_e 1)C_1$ whose external symbol e_e is at position 1, and then performing $X(e_e, \gamma_1)$ for replacing e_e by γ_1 to reach $W=(\gamma_1 1)C_1$. Finally, perform $F(C_1, \gamma_1)$ such that γ_1 can be corrected at last. In total, c heads apply R2; $m+2e-c$ symbols apply R1, and one edge is contributed by performing $X(e_e, \gamma_1)$. Thus, the length of the reconstructed $\pi(s_{2,1})$ is $c+m+2e+1 = d(S_{n,k}; \rho, \varepsilon)+1 = d(S_{n,k})+1$, since $\Delta_2=0$. Figure 7 illustrates the paths of Case 1.

According to Figure 7, the subpaths $[W-\varepsilon]_{\pi(s_{2,1})}$ and $(\rho_{s_{1,1}}-V)_{\pi(s_{1,1})}$ should be verified that they are vertex-disjoint to each other and vertex-disjoint to all α -paths, and $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$ as follows.

1. Symbol γ_1 is an inner external symbol only in each vertex of the subpath $[W-\varepsilon]_{\pi(s_{2,1})}$. Therefore, the

reconstructed subpath $[W-\varepsilon]_{\pi(s_{2,1})}$ is vertex-disjoint to all α -paths. By Corollary 2.2, the reconstructed subpath $(\rho_{s_{1,1}}-V)_{\pi(s_{1,1})}$ is vertex-disjoint to other α -paths, because s_{c,k_c} is prior to $s_{1,1}$ in the sequence of other α -paths. Hence, the reconstructed subpaths $[W-\varepsilon]_{\pi(s_{2,1})}$ and $(\rho_{s_{1,1}}-V)_{\pi(s_{1,1})}$ are vertex-disjoint to all α -paths other than subpath $[U-\varepsilon]_{\pi(\gamma_1)}$.

2. The subpath $(\rho_{s_{1,1}}-V)_{\pi(s_{2,1})}$ is vertex-disjoint to subpaths $[W-\varepsilon]_{\pi(s_{2,1})}$ and $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$, because symbol γ_1 is an inner external symbol in the subpaths $[W-\varepsilon]_{\pi(s_{2,1})}$ and $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$ but $(\rho_{s_{1,1}}-V)_{\pi(s_{2,1})}$ is not.
3. The subpaths $[W-\varepsilon]_{\pi(s_{2,1})}$ and $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$ are vertex-disjoint to each other, because C_c has already been corrected in $[W-\varepsilon]_{\pi(s_{2,1})}$ but $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$ has not yet.

Case 2. ($p_1 \neq 1$).

It is very similar to Case 1 of Refinement 2 if ρ has at least 2 cycles, or Case 2 of Refinement 2 if the external cycle is the only cycle which contains at least 3 external symbols, since γ_1 can be regarded as a fixed symbol in a pseudo position.

Based on the discussion above, Lemma 3.19 can be obtained as follows.

Lemma 3.19 There are $n-1$ vertex-disjoint paths from ρ to ε of length at most $d(S_{n,k})+1$ after reconstruction of Refinement 3.

3.2.2.4 Refinement 4: γ -Paths Reconstruction When $e'=2$

This refinement is designed for γ -paths when $e'=2$ as shown in the 4th row of Table 2. Two γ -paths should be reconstructed for shortening the paths. If the specific vertex ρ has at least 3 cycles, then the reconstruction of the first γ -path $\pi(\gamma_1)$ is similar to the Case 1 of Refinement 2. If the specific vertex ρ , whose C_1 is formed as $(m_1 e_1 \dots d_1 1)$, has at least 2 cycles and one of them is the external cycle $C_1 = (m_1 e_1 \dots d_1 1)$, then the reconstruction of the first γ -path $\pi(\gamma_1)$ is similar to the Case 2 of Refinement 2.

Notably, no β -paths are constructed in this situation, since no fixed symbols exist in ρ . Two cases of the refinement for reconstructing $\pi(\gamma_1)$ and $\pi(\gamma_2)$ are stated below.

Case 1. (ρ has at least 3 cycles).

Let ρ be $C_1 C_2 \dots C_c$, where $c \geq 3$. Reconstruct $\pi(\gamma_1)$ by performing $X(p_1, \gamma_1)$ and $F(C_2, \dots, C_c, \gamma_1, C_1)$. In total, one edge is contributed by performing $X(p_1, \gamma_1)$; $c-1$ heads apply rule R2, and $m+2e-(c-1)$ symbols apply rule R1. Referring to the 6th row of Table 2, $\Delta_2 = \Delta_3 = 1$ or 2, and the length of reconstructed $\pi(\gamma_1)$ is thus $1+2(c-1)+(m+2e-c+1) = c+m+2e = d(S_{n,k}; \rho, \varepsilon)+2 \leq d(S_{n,k})+1$. Figure 8 illustrates the reconstructed

path $\pi(\gamma_1)$. Reconstruct $\pi(\gamma_2)$ by performing $X(p_1, \gamma_2)$, $F(C_2, \gamma_2)$ and $F(C_1, C_3, \dots, C_c)$. In total, one edge is contributed by performing $X(p_1, \gamma_2)$; $c-1$ heads apply rule R2, and $m+2e-(c-1)$ symbols apply rule R1. Thus, the length of reconstructed $\pi(\gamma_2)$ is $1+2(c-1)+(m+2e-c+1) = c+m+2e = d(S_{n,k}; \rho, \varepsilon)+2 \leq d(S_{n,k})+1$, since $\Delta_2 = \Delta_3 = 1$ or 2. Figure 9 illustrates the reconstructed path $\pi(\gamma_2)$.

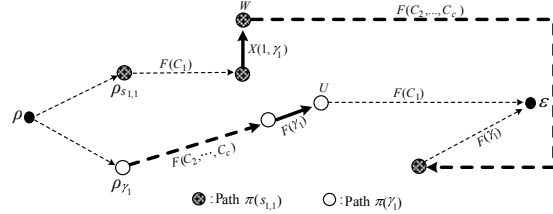


Figure 8. Case 1 of Refinement 4 for reconstructing the first γ -path $\pi(\gamma_1)$.

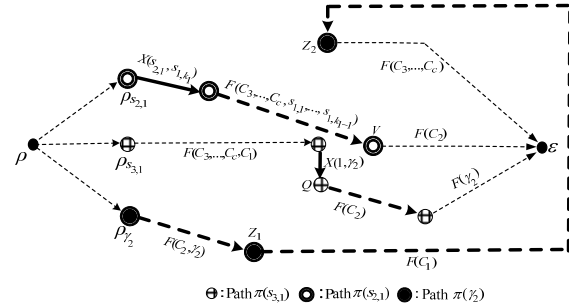


Figure 9. Case 1 of Refinement 4 for reconstructing the second γ -path $\pi(\gamma_2)$.

However, the reconstructed path $\pi(\gamma_1)$ and the α -path $\pi(s_{2,1})$ intersect at vertex $U=C_1$ as shown in Figure 8, because $\pi(s_{2,1})$ is constructed by $\langle C_2, \dots, C_c, C_1 \rangle$, and when $C_2, \dots, C_c$ are fixed, both paths contain vertex U . Hence, $\pi(s_{2,1})$ should be reconstructed to exclude vertex U . Moreover, the reconstructed path $\pi(\gamma_2)$ and the α -path $\pi(s_{1,1})$ intersect at vertex $Z_2=C_3 \dots C_c$ as shown in Figure 9, because $\pi(s_{1,1})$ is constructed by $\langle C_1, \dots, C_c \rangle$, and when the symbols before C_3 are fixed, both paths contain vertex Z_2 . Hence, $\pi(s_{1,1})$ should be reconstructed to exclude vertex Z_2 .

Reconstruct $\pi(s_{1,1})$ by performing $F(C_1)$, $X(1, \gamma_1)$, and then $F(C_2, \dots, C_c, \gamma_1)$ as shown in Figure 8. In total, one edge is contributed by performing $X(1, \gamma_1)$; $c-1$ heads apply rule R2, and $m+2e-(c-1)$ symbols apply rule R1. Thus, the length of the reconstructed $\pi(s_{1,1})$ is $1+2(c-1)+(m+2e-c+1) = c+m+2e = d(S_{n,k}; \rho, \varepsilon)+2 \leq d(S_{n,k})+1$, since $\Delta_2 = \Delta_3 = 1$ or 2.

Restated, symbol 1 is also represented by s_{1,k_1} in this case. Reconstruct $\pi(s_{2,1})$ by performing $X(p_{1,s_{2,1}})$ and $X(s_{2,1}, s_{1,k_1})$ such that $(s_{1,1} \dots s_{1,k_1-1} \langle C_2 \rangle)$ forms a new cycle, and then performing $F(C_3, \dots, C_c, s_{1,1}, \dots, s_{1,k_1-1}, C_2)$ to correct $c-1$ cycles as shown in Figure 9. In total, two edges are contributed by performing $X(p_{1,s_{2,1}})$ and $X(s_{2,1}, s_{1,k_1})$; $c-1$ heads apply rule R2,

and $m+2e-1-(c-1)$ symbols apply rule R1. Thus, the length of the reconstructed $\pi(s_{2,1})$ is $2+2(c-1)+(m+2e-c) = c+m+2e = (S_{n,k}; \rho, \varepsilon)+2 \leq d(S_{n,k})+1$, since $\Delta_2=\Delta_3=1$ or 2.

Unfortunately, the reconstructed path $\pi(s_{2,1})$ and the α -path $\pi(s_{3,1})$ intersect at vertex $V=C_2$ as shown in Figure 9, because $\pi(s_{3,1})$ is constructed by $\langle C_3, \dots, C_c, C_1, C_2 \rangle$, and when the symbols before C_2 are fixed, both paths contain vertex V . Hence, reconstruct $\pi(s_{3,1})$ by performing $F(C_3, C_4, \dots, C_c, C_1), X(1, \gamma_2)$ and $F(C_2, \gamma_2)$ as shown in Figure 9. In total, one edge is contributed by performing $X(1, \gamma_2)$; $c-1$ heads apply rule R2, and $m+2e-(c-1)$ symbols apply rule R1. Thus the length of the reconstructed $\pi(s_{3,1})$ is $1+2(c-1)+(m+2e-c+1) = c+m+2e = d(S_{n,k}; \rho, \varepsilon)+2 \leq d(S_{n,k})+1$, since $\Delta_2=\Delta_3=1$ or 2.

According to Figure 8 and Figure 9, five subpaths $[W-\varepsilon]_{\pi(s_{1,1})}$, $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$, $(\rho_{s_{2,1}}-V)_{\pi(s_{2,1})}$, $[Q-\varepsilon]_{\pi(s_{3,1})}$, and $(\rho_{\gamma_2}-Z_2)_{\pi(\gamma_2)}$ should be verified that they are vertex-disjoint to each other, and vertex-disjoint to $(\rho_{s_{1,1}}-W)_{\pi(s_{1,1})}$, $[U-\varepsilon]_{\pi(\gamma_1)}$, $[Z_2-\varepsilon]_{\pi(\gamma_2)}$, $[V-\varepsilon]_{\pi(s_{2,1})}$, and $(\rho_{s_{3,1}}-Q)_{\pi(s_{3,1})}$, and all α -paths other than $\pi(s_{1,1})$, $\pi(s_{2,1})$ and $\pi(s_{3,1})$ as follows.

1. Symbol γ_1 is only in each vertex of $[W-\varepsilon]_{\pi(s_{1,1})}$ and $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$. Symbol γ_2 is only in each vertex of $[Q-\varepsilon]_{\pi(s_{3,1})}$ and $(\rho_{\gamma_2}-Z_2)_{\pi(\gamma_2)}$. Therefore, $[W-\varepsilon]_{\pi(s_{1,1})}$, $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$, $[Q-\varepsilon]_{\pi(s_{3,1})}$ and $(\rho_{\gamma_2}-Z_2)_{\pi(\gamma_2)}$ are vertex-disjoint to all α -paths, and $[Q-\varepsilon]_{\pi(s_{3,1})}$ and $(\rho_{\gamma_2}-Z_2)_{\pi(\gamma_2)}$ are vertex-disjoint to $[W-\varepsilon]_{\pi(s_{1,1})}$ and $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$.
2. According to Corollary 2.2, $[Z_1-Z_2]_{\pi(\gamma_2)}$ is vertex-disjoint to all α -paths other than $\pi(s_{2,1})$, because $s_{2,1}$ is prior to s_{1,k_1} in the sequence of $(\rho_{\gamma_2}-Z_2)_{\pi(\gamma_2)}$ but in each sequence of all α -paths other than $\pi(s_{2,1})$ is not. Moreover, $[Z_1-Z_2]_{\pi(\gamma_2)}$ is also vertex-disjoint to $[U-\varepsilon]_{\pi(\gamma_1)}$, because C_c is fixed in $[U-\varepsilon]_{\pi(\gamma_1)}$ but $[Z_1-Z_2]_{\pi(\gamma_2)}$ is not.
3. After performing $X(s_{2,1}, s_{1,k_1}), (s_{2,1} \dots s_{2,k_2}, s_{1,1} \dots s_{1,k_1-1})$ forms a new cycle. Subpath $(\rho_{s_{2,1}}-V)_{\pi(s_{2,1})}$ and all α -paths are vertex-disjoint, since symbol $s_{2,1}$ is at position s_{1,k_1-1} for each vertex in $(\rho_{s_{2,1}}-V)_{\pi(s_{2,1})}$, but no vertices in a α -path have symbol $s_{2,1}$ residing at position s_{1,k_1-1} .
4. As mentioned above, $[W-\varepsilon]_{\pi(s_{1,1})}$, $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$, $[Q-\varepsilon]_{\pi(s_{3,1})}$, $(\rho_{\gamma_2}-Z_2)_{\pi(\gamma_2)}$ and $(\rho_{s_{2,1}}-V)_{\pi(s_{2,1})}$ are vertex-disjoint to $[U-\varepsilon]_{\pi(\gamma_1)}$ and all α -paths other than $\pi(s_{2,1})$.
5. Subpath $(\rho_{s_{2,1}}-V)_{\pi(s_{2,1})}$ is vertex-disjoint to $[W-\varepsilon]_{\pi(s_{1,1})}$, $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$, $[Q-\varepsilon]_{\pi(s_{3,1})}$ and $(\rho_{\gamma_2}-Z_2)_{\pi(\gamma_2)}$, because symbols γ_1 and γ_2 are not in each vertex of $(\rho_{s_{2,1}}-V)_{\pi(s_{2,1})}$. Moreover, $(\rho_{s_{2,1}}-V)_{\pi(s_{2,1})}$ and $[Z_1-Z_2]_{\pi(\gamma_2)}$ are also vertex-disjoint, because C_2 is fixed in $[Z_1-Z_2]_{\pi(\gamma_2)}$, but $(\rho_{s_{2,1}}-V)_{\pi(s_{2,1})}$ is not. Therefore, $(\rho_{s_{2,1}}-V)_{\pi(s_{2,1})}$ is vertex-disjoint to $[W-\varepsilon]_{\pi(s_{1,1})}$, $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$, $[Q-\varepsilon]_{\pi(s_{3,1})}$ and $(\rho_{\gamma_2}-Z_2)_{\pi(\gamma_2)}$.
6. Subpaths $[Q-\varepsilon]_{\pi(s_{3,1})}$ and $(\rho_{\gamma_2}-Z_2)_{\pi(\gamma_2)}$ are vertex-disjoint to each other, because C_c is fixed in $[Q-\varepsilon]_{\pi(s_{3,1})}$ but $(\rho_{\gamma_2}-Z_2)_{\pi(\gamma_2)}$ is not.

7. Subpaths $[W-\varepsilon]_{\pi(s_{1,1})}$ and $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$ are vertex-disjoint to each other, because C_1 is fixed in $[W-\varepsilon]_{\pi(s_{1,1})}$ but $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$ is not.

Case 2. (ρ has at least 2 cycles and one of them is the external cycle $C_1 = (m_1 e_1 \dots d_1 1)$).

Let $\rho = C_1 C_2 \dots C_c$, where $c \geq 2$ and $C_1 = (s_{1,1} s_{1,2} \dots s_{1,k_1-1} s_{1,k_1}) = (m_1 e_1 \dots d_1 1)$. Reconstruct $\pi(\gamma_1)$ by performing $X(m_1, \gamma_1), F(C_2, \dots, C_c), X(\gamma_1, e_1), F(e_1, m_1, \gamma_1)$ and $F(s_{1,3}, \dots, d_1)$. In total, 2 edges are contributed by performing $X(m_1, \gamma_1)$ and $X(\gamma_1, e_1)$; $c-1$ heads apply rule R2, and $m+2e-(c-1)$ symbols apply rule R1. Referring to the 7th row of Table 2, $\Delta_2=\Delta_3=2$ and the length of reconstructed $\pi(\gamma_1)$ is thus $2+2(c-1)+(m+2e-c+1) = c+m+2e+1 = d(S_{n,k}; \rho, \varepsilon)+3 = d(S_{n,k})+1$. Reconstruct $\pi(\gamma_2)$ by performing $X(m_1, \gamma_2), X(\gamma_2, e_1)$ and $F(e_1, m_1, \gamma_2, s_{1,3}, \dots, d_1, 1, C_2, \dots, C_c)$. In total, 2 edges are contributed by performing $X(m_1, \gamma_2)$ and $X(\gamma_2, e_1)$; $c-1$ heads apply rule R2, and $m+2e-(c-1)$ symbols apply rule R1. Thus, the length of reconstructed $\pi(\gamma_2)$ is $2+2(c-1)+(m+2e-c+1) = c+m+2e+1 = d(S_{n,k}; \rho, \varepsilon)+3 = d(S_{n,k})+1$, since $\Delta_2=\Delta_3=2$. Figure 10 illustrates the reconstructed paths $\pi(\gamma_1)$ and $\pi(\gamma_2)$.

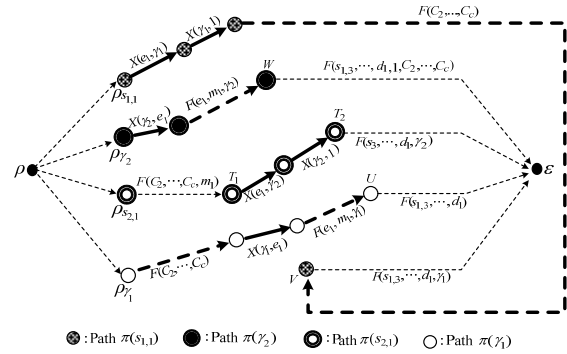


Figure 10. Case 2 of Refinement 4.

However, the reconstructed path $\pi(\gamma_1)$ and α -path $\pi(s_{2,1})$ intersect at vertex $U=(s_{1,3}, \dots, d_1 1)$ as shown in Figure 10, because $\pi(s_{2,1})$ is constructed by $\langle C_2, \dots, C_c, C_1 \rangle$, and when $C_2, \dots, C_c, m_1$ and e_1 are fixed, both paths contain vertex U . Moreover, the reconstructed path $\pi(\gamma_2)$ and α -path $\pi(s_{1,1})$ also intersect at vertex $W=(s_{1,3}, \dots, d_1 1)C_2 \dots C_c$ as shown in Figure 10, because $\pi(s_{1,1})$ is constructed by $\langle C_1, \dots, C_c \rangle$, and when m_1 and e_1 are fixed, both paths contain vertex W . Hence, reconstruct $\pi(s_{2,1})$ by performing $F(C_2, \dots, C_c, m_1), X(e_1, \gamma_2), X(\gamma_2, 1)$ and $F(s_{1,3}, \dots, d_1, \gamma_2)$, and reconstruct $\pi(s_{1,1})$ by performing $F(m_1), X(e_1, \gamma_1), X(\gamma_1, 1), F(C_2, \dots, C_c)$ and $F(s_{1,3}, \dots, d_1, \gamma_1)$. Notably, $X(e_1, \gamma_1)$ (or $X(e_1, \gamma_2)$) is used to replace e_1 by γ_1 (or γ_2) such that e_1 is removed and no longer needed to be corrected. In total, 2 edges are contributed by performing $X(e_1, \gamma_1)$ and $X(\gamma_1, 1)$ ($X(e_1, \gamma_2)$ and $X(\gamma_2, 1)$); c heads apply rule R2, and $m+2e-1-c$ symbols apply rule R1. Thus, the length of the reconstructed α -path $\pi(s_{1,1})$ ($\pi(s_{2,1})$) is

$2+2c+(m+2e-1-c)=c+m+2e+1=d(S_{n,k}; \rho, \varepsilon)+3=d(S_{n,k})+1$. All reconstructed paths are shown in Figure 10.

According to Figure 10, four subpaths $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$, $(T_1-T_2)_{\pi(s_{2,1})}$, $(\rho_{\gamma_2}-W)_{\pi(\gamma_2)}$ and $(\rho_{s_{1,1}}-V)_{\pi(s_{1,1})}$ should be verified that they are vertex-disjoint to each other and vertex-disjoint to all α -paths, and vertex-disjoint to $[V-\varepsilon]_{\pi(s_{1,1})}$ and $[T_2-\varepsilon]_{\pi(s_{2,1})}$ which are the subpaths of the original γ -paths $\pi(\gamma_1)$ and $\pi(\gamma_2)$, respectively.

1. Symbol γ_1 is only in each vertex of $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$ and $(\rho_{s_{1,1}}-\varepsilon)_{\pi(s_{1,1})}$. Symbol γ_2 is only in each vertex of $(T_1-T_2)_{\pi(s_{2,1})}$ and $(\rho_{\gamma_2}-W)_{\pi(\gamma_2)}$. Therefore, $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$, $(\rho_{s_{1,1}}-\varepsilon)_{\pi(s_{1,1})}$, $(T_1-T_2)_{\pi(s_{2,1})}$ and $(\rho_{\gamma_2}-W)_{\pi(\gamma_2)}$ are vertex-disjoint to all α -paths, and $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$ and $(\rho_{s_{1,1}}-\varepsilon)_{\pi(s_{1,1})}$ are vertex-disjoint to $(T_1-T_2)_{\pi(s_{2,1})}$ and $(\rho_{\gamma_2}-W)_{\pi(\gamma_2)}$.
2. According to Corollary 2.2, $(\rho_{s_{1,1}}-V)_{\pi(s_{1,1})}$ and $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$ are vertex-disjoint to each other, because the symbol s_{c,k_c} is prior to m_1 in the sequence of $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$.
3. Subpaths $(T_1-T_2)_{\pi(s_{2,1})}$ and $(\rho_{\gamma_2}-W)_{\pi(\gamma_2)}$ are vertex-disjoint to each other, because C_2 is fixed in $(T_1-T_2)_{\pi(s_{2,1})}$ but $(\rho_{\gamma_2}-W)_{\pi(\gamma_2)}$ is not.
4. Subpaths $[V-\varepsilon]_{\pi(s_{1,1})}$ and $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$ are vertex-disjoint to each other, because symbol γ_1 is at position $d_1 (=s_{1,k_1-1})$ for each vertex in $[V-\varepsilon]_{\pi(s_{1,1})}$ but each vertex in $(\rho_{\gamma_1}-U)_{\pi(\gamma_1)}$ is not.
5. Subpaths $[T_2-\varepsilon]_{\pi(s_{2,1})}$ and $(\rho_{\gamma_2}-W)_{\pi(\gamma_2)}$ are vertex-disjoint to each other, because C_2 is fixed in $[T_2-\varepsilon]_{\pi(s_{2,1})}$ but in $(\rho_{\gamma_2}-W)_{\pi(\gamma_2)}$ is not.

According to the discussion above, Lemma 3.20 can be obtained as follows.

Lemma 3.20 There are $n-1$ vertex-disjoint paths from ρ to ε of length at most $d(S_{n,k})+1$ after reconstruction of Refinement 4.

Four refinements are provided to reconstruct the paths of length at most $d(S_{n,k})+1$ for some specific ρ s as shown in Table 2. Lemmas 3.17 to 3.20 indicate that an upper bound of $d_{n-1}(S_{n,k})$, for $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n \geq 7$, can be derived as follows.

Lemma 3.21 If $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n \geq 7$, then $d_{n-1}(S_{n,k}) \leq d(S_{n,k})+1$.

According to Lemmas 3.8, 3.12, and 3.21, $d_{n-1}(S_{n,k})$ can be determined and stated in Theorem 3.22.

Theorem 3.22. $d_{n-1}(S_{n,k}) = d(S_{n,k})+2$ ($d_{n-1}(S_{n,k}) = d(S_{n,k})+1$) if $2 \leq k \leq \lfloor n/2 \rfloor$, or $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n < 7$ (if $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n \geq 7$).

4 Conclusion

This work shows that $d_k(G)$ and $D_k(G)$ are bounded below by $d(G)+1$ for a regular graph with a connectivity $\kappa \geq 2$ and $d(G) > 2$. Since an $S_{n,k}$ is a regular graph with a connectivity $\kappa \geq 2$ and $d(S_{n,k}) > 2$,

the lower bounds of $d_{n-1}(S_{n,k})$ and $D_{n-1}(S_{n,k})$ are thus $d(S_{n,k})+1$. Additionally, a better lower bound of $d_{n-1}(S_{n,k})$ for $S_{5,3}$ and $S_{6,4}$ is $d(S_{n,k})+2$ and obtained by constructing a best container from some vertex ρ to ε in the specific $S_{n,k}$. Moreover, given any vertex ρ in a $S_{n,k}$ with $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n \notin \{5, 6\}$, a new $C_{n-1}(\rho, \varepsilon)$ with a length at most $d(S_{n,k})+1$ can be built by modifying the $C_{n-1}(\rho, \varepsilon)$ constructed by the one-to-one routing algorithm which proposed by Lin *et al.* These measurement results demonstrate that $d_{n-1}(S_{n,k})=d(S_{n,k})+2$ if $2 \leq k \leq \lfloor n/2 \rfloor$, $d_4(S_{5,3})=d(S_{5,3})+2$, $d_5(S_{6,4})=d(S_{6,4})+2$, and $d_{n-1}(S_{n,k})=d(S_{n,k})+1$ if $\lfloor n/2 \rfloor + 1 \leq k \leq n-2$ and $n \notin \{5, 6\}$.

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